

EFS-02 Summary Document

Overview:

This document provides a summary of the EFS-02 lecture on Statistical Arbitrage. The lecture covers the following topics:

- Types of statistical arbitrage
- Stationarity and statistical method to test it such as ADF Test
- Bollinger bands strategy
- Pair trading
- Cointegration vs correlation
- Best markets to pair trade
- Risk management practices

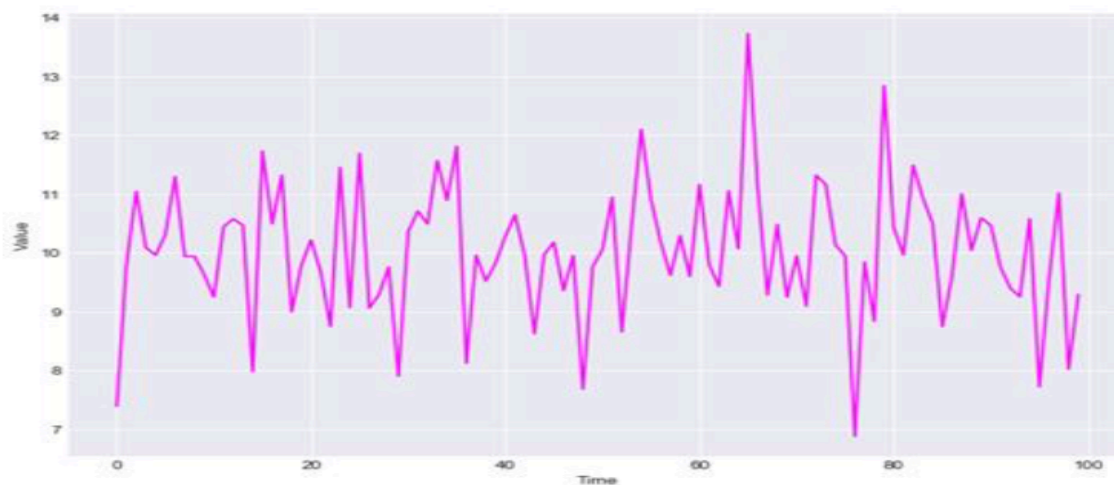
Statistical arbitrage

Statistical arbitrage is a type of trading strategy which uses the deviation of the price of an instrument relative to another instrument(s) and tries to profit from the deviation.

Stationarity

A series is considered stationary if it has a constant mean and variance over time.

The following snapshot shows an example of a stationary series.



Test for stationarity

We can use the Augmented Dickey-Fuller (ADF) test to check for stationarity in a given time series.

Why not just backtest the series to check for stationarity? Why do we need a statistical test to check for it?

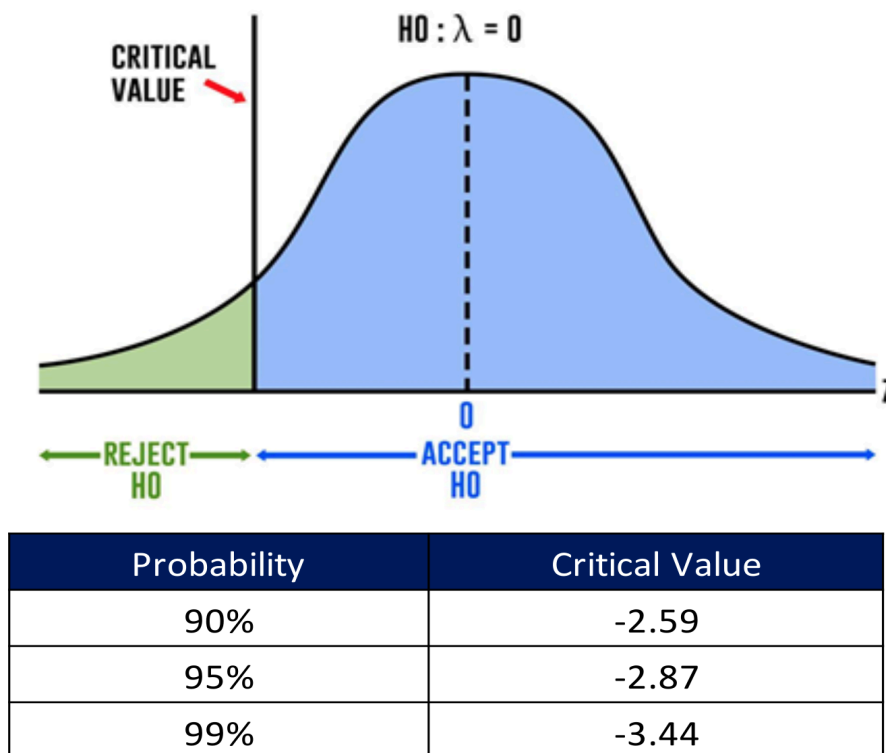
1. The backtest output depends on the input parameters, whereas a statistical test incorporates all the points.
2. Statistical tests are faster.

ADF test

The Augmented Dickey-Fuller (ADF) test is one of the most commonly used tests for stationarity. It has a null and an alternative hypothesis. The null hypothesis is that the series is non-stationary.

Here's an example to get an understanding of how to interpret the results of this test:

The following snapshot shows the critical values and their respective confidence intervals



For example:

- If t-stat < -2.59, then the series is stationary with 90% certainty
- If t-stat < -2.87, then the series is stationary with 95% certainty
- If t-stat < -3.44, then the series is stationary with 99% certainty

ADF test can be implemented in python using `adfuller()` function from the `statsmodels.tsa.stattools` package

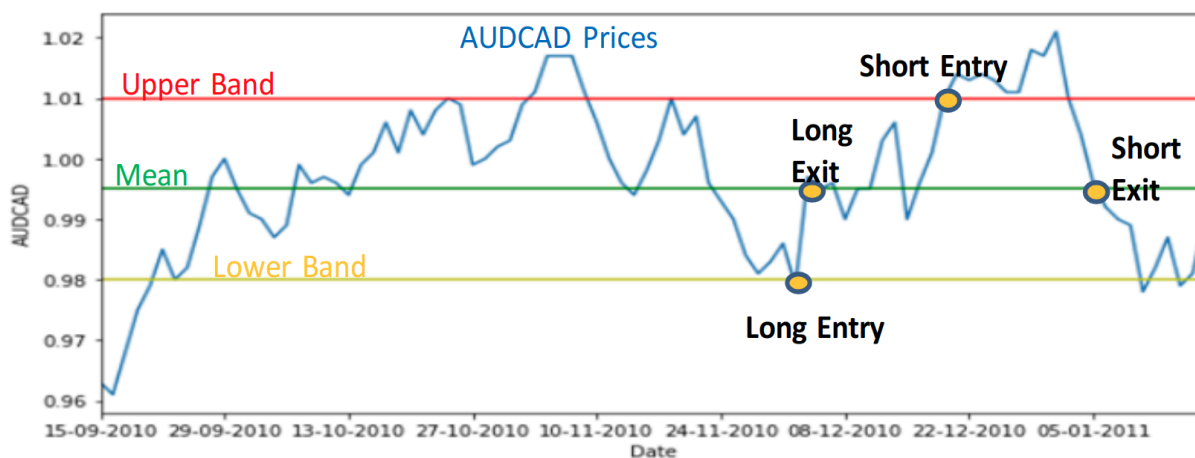
Bollinger bands strategy

Bollinger bands are bands plotted above and below a particular standard deviation from the simple moving average. The line plotted above the moving average line is the upper band, and the one below the moving average line is called the lower band.

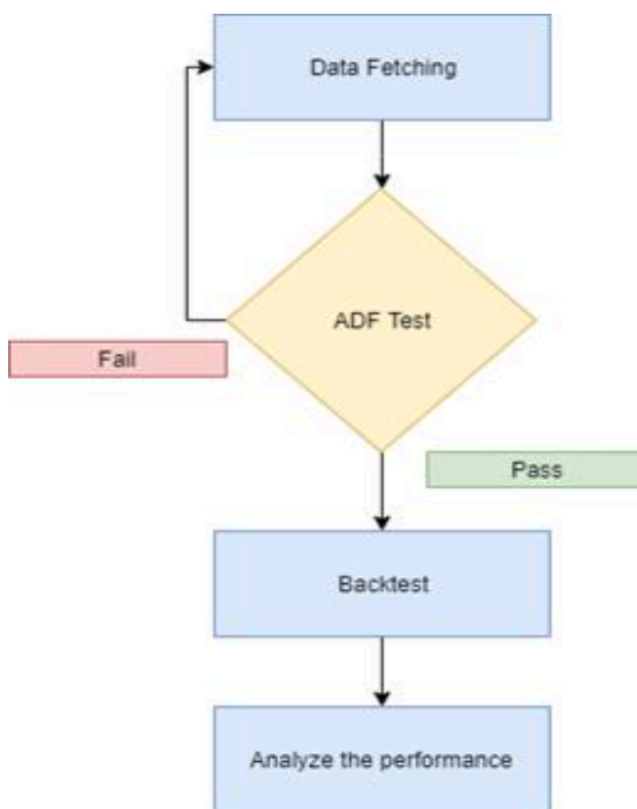
Steps for Mean Reversion Strategy

1. Read the close price data from the CSV file (or any other file format)
2. Calculate moving average and moving standard deviation (SD) of the series
3. Calculate Bollinger bands, i.e. upper and lower bands
4. Buy and sell positions
 - Long entry: If close price < lower band
 - Long exit: If close price > moving average
 - Short entry: If close price > upper band
 - Short exit: If close price < moving average

The following snapshot shows an example of the mean reversion strategy using Bollinger bands for the instrument AUDCAD (Australian and Canadian dollar exchange rate)



Summarising mean reversion strategy



Pair Trading

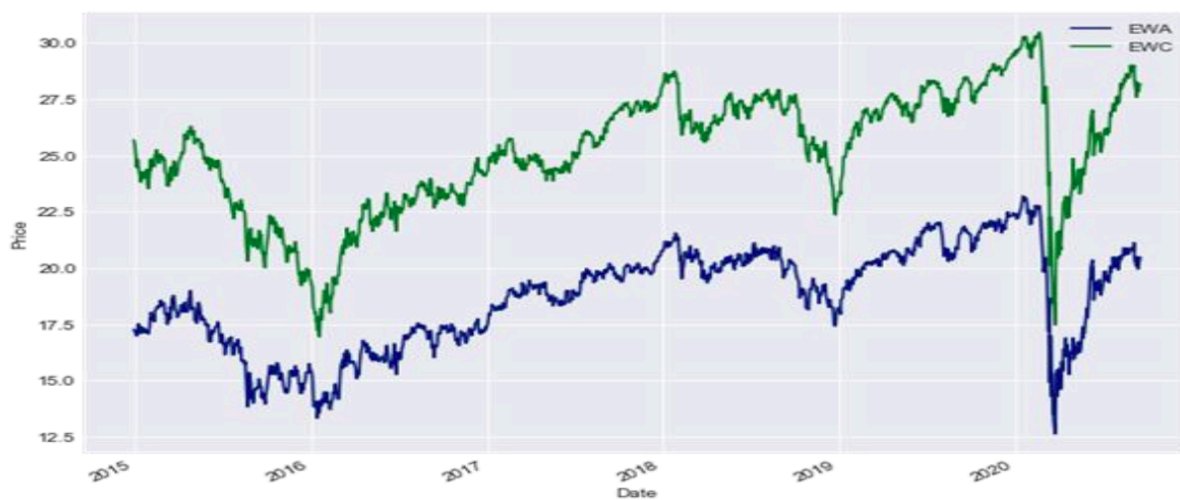
Pair trading uses relative prices of 2 instruments to take long and short positions depending on the deviation of the spread (difference of the price of 2 instruments) from its mean. There are different versions of pair trading with more than two instruments.

For example - Triplets use three instruments for statistical arbitrage.

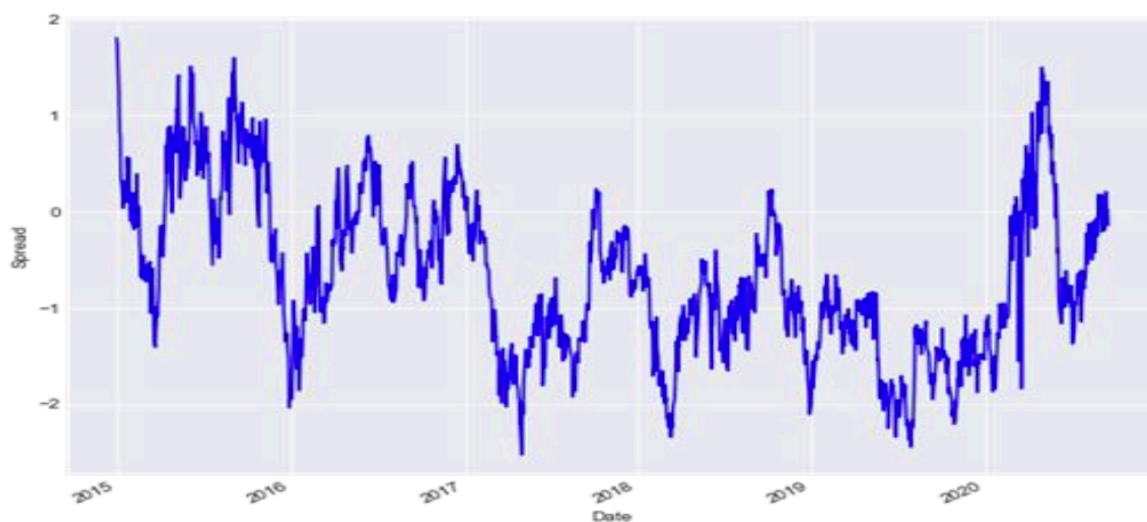
Co-integration

When the time series of the spread (the difference in prices of the two instruments) is stationary, consider the time series of the two instruments involved are said to be co-integrated.

The following snapshot shows an example of 2 co-integrated series time series.



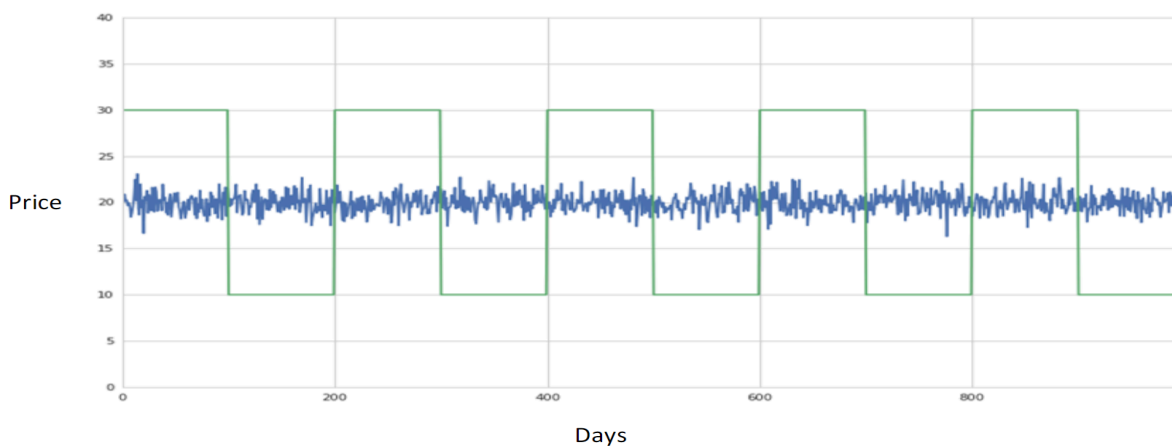
The following snapshot shows the spread of the two co-integrated series presented above.



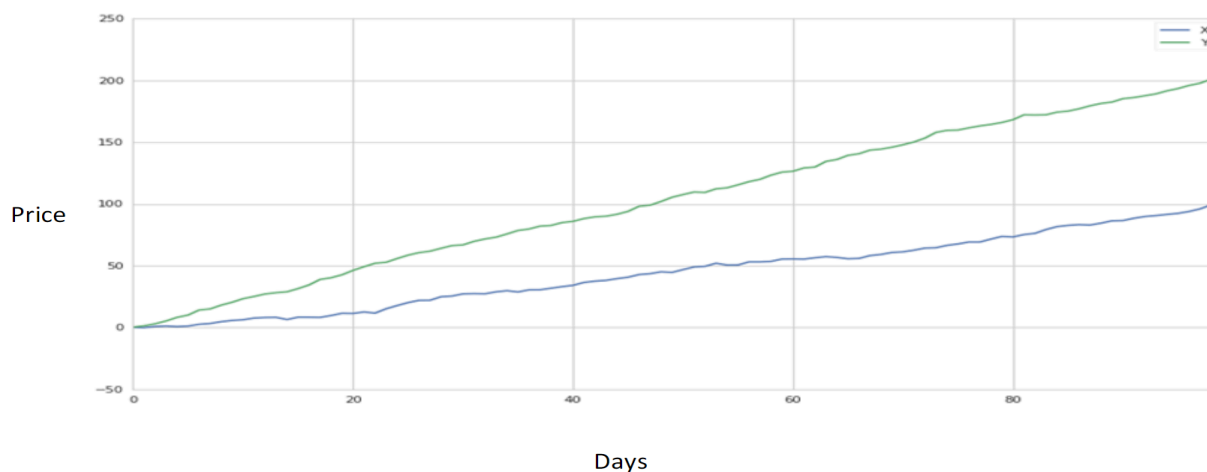
Cointegration vs correlation

- Cointegration is the stationarity of the spread of 2 instruments
- Correlation is the movement of 2 instruments in the same direction.

The following snapshot shows the example of co-integrated instruments which are not correlated.



The following snapshot shows the example of correlated instruments which are not co-integrated.



Why does pair trading work?

1. Cointegrated pairs are assumed to be mean reverting in nature
2. Spread is likely to revert to mean if it diverges
3. The spread of instruments from the same sector is unlikely to diverge from the mean significantly.

Mean Reversion Strategy

- Buy the spread when it goes lower than the lower band
- Sell the spread when it goes above the mean/upper band

Best markets to pair trade

Pair selection criteria

- Qualitative selection (ETF and currency)
 - Exposed to same economic factors

- Qualitative selection (Stocks)
 - Same Sector
 - Similar Market Capitalization
 - Similar Ratios

Co-integration check

- ADF test

Risk Management

Why is risk management required?

- Statistical arbitrage is not a risk-free strategy.
- An event in a particular instrument can trigger extreme movement in the spread. Risk management is required once the mean-reverting behaviour of the spread is invalidated.

How to do risk management?

- Define stop-loss and profit mechanisms before entering a trade.
- Diversify funds in different pair trading strategies.
- Use a combination of pair trading and momentum trading strategies.

Lessons from LTCM

- Over-reliance on backtesting - We assume the past will be similar to the future. Still, we should be wary of this assumption and build our strategies, considering the possibility of this assumption being invalidated.
- High leverage - Taking leverage without considering its risk can lead to catastrophic losses and bankruptcy.

Additional resources

- [Arbitrage Strategies: Understanding Working of Statistical Arbitrage](#)
- Ernest P. Chan "Algorithmic trading winning strategies and their rationale."
- Gatev, Evan, William N. Goetzmann, and K. Geert Rouwenhorst, "Pairs Trading: Performance of a Relative-Value Arbitrage Rule," Review of Financial Studies (2006): 797-827
- [Statistical Arbitrage in the U.S. Equities Market](#)
- Github links
 - <https://github.com/QuantInsti/EPAT/blob/master/Statistical%20Arbitrage/optimization.py>
 - https://github.com/QuantInsti/EPAT/blob/master/Statistical%20Arbitrage/half_life.py